

Responsibility of Fishermen to Fisheries Resource Management at Kiew Lom Dam, Lampang Province

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ABSTRACT A study of fishermen responsibilities in the Kiew Lom Dam Lampang province with regards to Fisheries Resource Management. This study aims to evaluate the level of responsibility and factors affecting the responsibility of fishermen in the management of fisheries resources at the Kiew Lom Dam, Lampang. In conducting this research, data was collected from a total of 75 fishermen using structural interviews. Most fishermen participating in the study were male (81.33 %), most were over the age of 46 (80.00 %), most graduated primary school (53.33 %), almost all of the fishermen were married (98.67 %), almost all of the fishermen were locals in Lampang province (98.67 %), and most had no official position in the community (92.00 %). Fisheries primarily provide a supplemental occupation (82.67 %), with only a small minority of fishermen working in fisheries as their primary occupation (9.33 %). The majority of fishermen have a high level of knowledge of resources and measures to manage fisheries (82.67%). Knowledge of resources and resource management is primarily from attendance and listening (72.72 %) and government provided support. Research shows that 90.67% of fishermen have supported fishing resource management activities from government agencies. Research also has found that more than half of fishermen have a very high level of responsibility for themselves (54.67%), and most fishermen have a high level of social responsibility (66.67%). Lampang province maintains statistically significant fisheries resource management ($P < 0.05$) which includes resource knowledge and fisheries resource management measures, community group attendance, and support for fisheries resource management activities from government agencies. Therefore, it should encourage more local fishermen to have knowledge of resources and measures to manage fisheries resources. Research should also consider the social support system that exists, such as the government holding meetings of fishermen groups to provide resource knowledge and measures to manage fisheries resources, or the holding of forums to allow fishermen to discuss resource management in the area discuss responsibilities for managing fishing resources.

Keywords: Readiness, Fisheries Management and Kiew Lom Dam.

I. INTRODUCTION

Preface

Kiew Lom Dam is a large concrete dam built with Gravity Dam reinforced concrete, holding 112 million cubic meters of water. The average annual flow of water flows into the basin is 578 million cubic meters. The Kiew Lom Dam has an average water reception area of 1,425 sq km. Kiew Lom Dam is a multipurpose dam used for irrigation. The Wang River barrier was built at Ban Lang Subdistrict, Mueang District, Lampang Province, which is a dam under the supervision of the Department of Irrigation. The Kiew Lom Dam also benefits a large number of people in nearby communities, benefits include a water source for the farmland of the Mae Wang Irrigation Project and the former Kiew Lom Irrigation Project, a water source for current and future consumer and industrial water uses, and reduced flooding at the end of the dam, especially in Muang District, Lampang Province. Water is also stored for use in some Mae Mau power plant activities and for future recreational use and as a new tourist attraction in Lampang province. Kiew Lom Dam is also a fisheries-based dam that is a breeding ground for breeding freshwater species, making it an important source of protein for locals, as well as a fishing ground for people in the community around Kiew Lom Dam. These factors impact the distribution of income in society (Royal Irrigation Department, 2019).

In 2012 there were 89 fishermen (Ong-art and et.al, 2012). But currently, there are only 75 fishermen due to lower aquatic problems, higher fishing costs, as well as uncertainty. As a result, fishermen have switched to other professions, such as trading and regular employment. In addition to this, there is a problem with fishermen lacking integration to add value to aquatic animals (Karn and et.al, 2020) under the conditions of development in order to solve fisheries management problems. Fishing resources around Kiew Lom Dam are also trending downward. As can be seen in the decreasing number of aquatic animals. Although the government has taken steps to manage fishing resources by issuing fishing regulations and measures, as well as issuing fisheries decrees, the public must ultimately participate in fisheries management by increasing fishing responsibilities for both society and for themselves.

This study evaluates the responsibility of fishermen at Kiew Lom Dam Lampang province and considers the level of responsibility and factors affecting fishermen's responsibility for managing fisheries resources at Kiew Lom Dam in order to achieve sustainable use of fishing resources. At the same time, government agencies can use this information on the fishing development of local fishermen to further align with the rules or regulations that will follow in the future.

II. EQUIPMENT AND METHODS

Equipment

1. Structure Questionnaire
2. Stationery Equipment
3. PC

III. RESEARCH METHODOLOGY

The study used personal structured interviews to interview a sample of fishing households living in and around Kiew Lom Dam Reservoir in two districts: Chae Hom District and Mueang District, Lampang Province, from October 2019 to September 2020.

Section 1 Social Basics, Fisherman's Economy at Kiew Lom Dam

Section 2 Knowledge of Resources and Measures for Fisheries Management

Section 3 Community Group Attendance

Section 4 Supporting Fisheries Resource Management Activities from Government Agencies

Section 5 Fisherman Responsibilities for Managing Fisheries Resources at Kiew Lom Dam

Quality check of the tools used to consolidate data

The validity test of the design of this interview will be based on relevant ideas and research results. To design the interview, then use the interview to present the experts to validate content validity, and then revise the interview to provide the correct content. Accuracy can be used to measure specific areas of focus.

Populations and samples

Population The population of this study was 75 fishermen living in fishing communities around Kiew Lom Dam, Chae-hom district and Mueang District, Lampang Province.

Sample methodology. The sampling method uses cluster sampling methods by simple sampling in villages around Kiew Lom Dam, so that randomly selected households represent the fishermen around Kiew Lom Dam. The number of samples represent a confidence level of 95 % using the Yamane formula (1973) as follows:

$$\text{Formula } n = \frac{N}{1+N(e^2)}$$

n = Number of samples
N = Population 75
e = The error proportion cannot exceed the specified value (0.05).

$$\text{From formula } \frac{75}{1+75(0.05^2)}$$

=63.16 Examples
=64 examples

Because the number of calculated figures is close to the population, the researchers collected data from a total of 75.

Data Collection

Data was collected by interviewing the population with structured interviews from October 2019 to September 2020.

Analysis of data and statistics used

1. Basic social and economic information: gender, age, education level Marital status, social status, fishing, calculating averages, %ages, and displayed as an explanation table, using ready-made programs for statistical analysis.

2. Testing the relationship between variables between fishermen's responsibilities to manage variable fisheries resources, including resource knowledge levels and measures for fisheries management, and support levels of fisheries resource management activities from government agencies. Analyze variable relationships with Pearson correlation coefficients (Pearson's correlation coefficient). For group meetings in the community, analyze the differences between attendees and non-attendance with t-test statistics at a significant level of 0.05.

IV. RESULT

1. Social Basics of the fisherman Economy at Kiew Lom Dam

For the study of basic social information, the fishermen's economy at Kiew Lom Fisherman Dam found that 81.33 % of the same as males, 18.67 % were female, and fishermen' ages were found to be 53 years old, with most fishermen over the age of 46 (80.00 %) aged 36-45, 10.67 %. 25 years, with 5.33 % of the education level, found that fishermen graduated the most primary school (65.33 %), second only to middle school (24.00 %), with only 4.00 % having a bachelor's degree in marital status.

In terms of social status, almost all fishermen have no social position (92.00 %), but there are still some of them as village committees (4.00 %), municipal members (2.67 %) and group presidents (1.33 %) (Table 5), in the domiciles of fishermen, found that most of them are domiciled in Lampang province (98.67 %), with only 1.33 % of them being outside the area, with relocated from Phayao province (98.67 %).

In terms of fishing, only 9.33 % of fishermen fishing is the main occupation. Most of them are fishing as secondary occupations. It is 82.67% and fishing for consumption is 8.00 % (Table1).

Table 1. Background information regarding Fisherman Economy at Kiew Lom Dam. (n=75)

General Information		Number (case)	Percentage
Gender	Male	61	81.33
	Female	14	18.67
Age	Below 18 years	1	1.33
	18-25 years	4	5.33
	26-35 years	2	2.67
	36-45 years	8	10.67
	46 years and over	60	80.00
Education level	Elementary School	49	65.33
	Secondary education	18	24.00
	Vocational Certificate	5	6.67
	Bachelor's Degree	3	4.00
Marital Status	Single	1	1.33
	Married	74	98.67
Official position	No social position	69	92
	Municipal Members	2	2.67
	Board Member	3	4
	Group President	1	1.33
Place of Origin	Local	74	98.67
	Phayao	1	1.33
Work Status as Fisherman	Primary Occupation	7	9.33
	Secondary Occupation	62	82.67
	Part Time Occupation	6	8.00

2. Knowledge of resources and measures in fisheries management

In resource knowledge, all fishermen were told that the most common aquatic animals in fishing are fish in the group, and the network is the most fishing tool in the area (100.00% equal). The most wrong part of the fisherman's response is that the reduced amount of aquatic animals has caused fishermen to give up their careers. In regards to fisheries management measures, it was found that almost all fishermen knew that according to the Fisheries Decree 2015, fishermen must have a fishing permit (93.33%) and knew that no one would be allowed to fish during freshwater

season with eggs in time and area. From June 1 to August 31, 2020 (88.00%), the most wrong fisherman's response was "local fishing", meaning that fishing in coastal sea zones, whether using fishing boats or using tools without fishing boats, is not commercial fisheries (16.00%) (Table 2).

The level of resource knowledge and measures in fisheries management showed that the majority of fishermen had a high level of knowledge of fisheries management resources and measures (82.67%) and a moderate 10.66%, with only 6.67% with low knowledge (Table 3).

Table 2. The percentage of Questions on Resource Knowledge Issues and Fisheries Management Measures. (n=75)

Knowledge	Correct (%)	Incorrect (%)
Fisheries Resources		
In the last 5 years, fishing resources at Kiew Lom Dam have decreased/	93.33	6.67
The most common aquatic creatures in fishing are fish in the group.	100.00	0.00
Summer is the highest time fishermen have achieved their output in years.	82.67	17.33
Networking is the most common fishing tool in the area.	100.00	0.00
The reduced amount of aquatic animals has caused fishermen to give up their careers.	72.00	28.00
Fisheries Management Measures		
According to the Fisheries Act 2015, "folk fishing" means fishing in coastal sea zones, whether using fishing boats or using tools without fishing boats.	84.00	16.00
According to the Fisheries Act 2015, fishermen must have a fishing permit.	93.33	6.67
It is forbidden for anyone to fish in freshwater season with eggs from June 1 to August 31, 2020.	88.00	12.00
Local fishing community organizations must be registered in accordance with the Fisheries Act 2015.	53.33	46.67

Table 3. Resource Knowledge and Fisherman Management Measures at Kiew Lom Dam (n=75)

Knowledge Level	Amount (case)	Percentage
Low (< 50 per cent)	5	6.67
Moderate (50-75 %)	8	10.66
Advanced (> 75 %)	62	82.67

3. Attending group meetings in the community

Group meetings in fishermen's communities were found to be mostly attended and listened to. A small minority participated to suggest feedback (16.36%) and participated in decisions (10.92%) (Table 4).

Table 4. The percentage of Group Participation Scores in Fishermen's Communities. (n=75)

Participation	Number (case)	Percentage
Do not participate	20	26.67
Participate, but listen only	40	72.72
Participate and make suggestions	9	16.36
Participate in the decision	6	10.92

4. Supporting fisheries resource management activities from government agencies

According to interviews with fishermen, almost all fishermen have commented that fisherman resource management activities from government agencies (93.33%) are supported by agencies that support fisheries resource management activities, namely the Department of Fisheries, but also supported by local authorities, including municipalities and district administrations (5.34%), and only one private agency has come to support fisheries resource management activities (1.33%) (Tarong 5).

Social support level

As for the level of support for fisheries resource management activities from government agencies, almost all fishermen have 90.67% support for fisheries resource management activities from government agencies and moderately high levels of 9.33% (Table 6).

Table 5. Supporting Fisheries Resource Management Activities from Government Agencies (n=75)

Government support	Number (case)	Percentage
Department of Fisheries	70	93.33
Local Authority	4	5.34
Private Agencies	1	1.33

Table 6. Level of Support for Fisheries Resource Management Activities from Government Agencies. (n=75)

Social support level	Number (Cases)	Percentage
Low level	0	0.00
Moderate level	7	9.33
High Level	68	90.67

5. Fisherman's Responsibility for Managing Fisheries Resources at Kiew Lom Dam

From Table 7, it is found that the responsibility for oneself in knowing the duties of good citizens in Thai society is found. It found that the most pragmatic issue for fishermen was "good membership for families" (93.33%), second only to "fishing integrity" (86.67%), while the issue of the most common to non-practical fishermen (less than 50 % of practice levels) was the issue of "giving up personal interests for others or society as a whole" (14.00%).

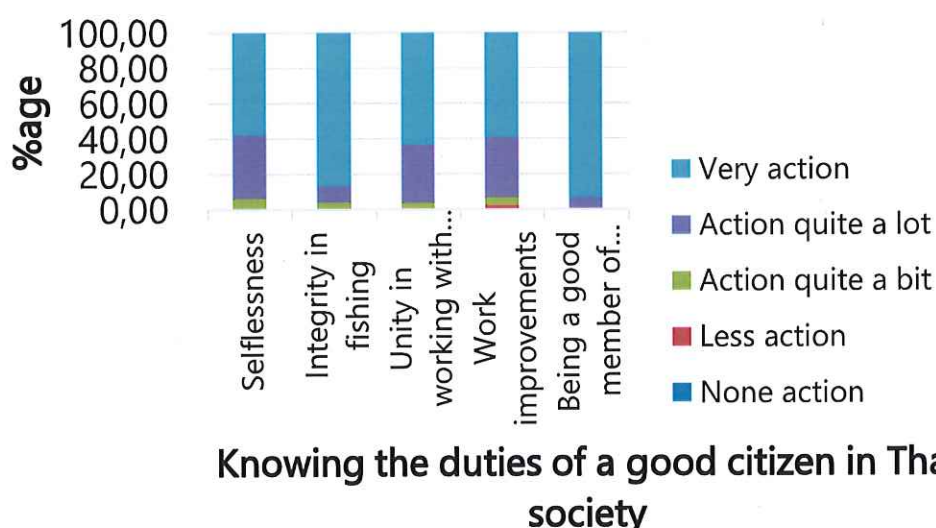


Figure 1. Self-responsibility in knowing the duties of a good citizen in Thai society
Social Compliance Self-Responsibility

It was found that the most pragmatic aspects of fishermen were "compliance with the prohibition of non-fishing in the red tide season" (93.33%), second only to the issue of "compliance with the number and type of fishing tools allowed in the license" and "controlling the use of natural resources and the environment in accordance with the regulations" (73.33%), but there were also issues that most fishermen representatives did not practice, the issue of "creating a fishing notebook, which must have the type and quantity of caught aquatic animals" (75.40%) and 14% of fishermen not to practice the issue of "personal income taxation."



Figure 2. Self-responsibility in social compliance

For self-responsibility in terms of self-reliance based on sufficiency economy. It found that the majority of fishermen were most practicing the issue of "proper use of resources according to their own force" (80.00%), second only to the issue of "resource use based on resource characteristics without damaging the natural system" (53.33%). There are some areas where more than a third of fishermen have set guidelines for managing fishing risks, such as finding alternative occupations (39.10%).

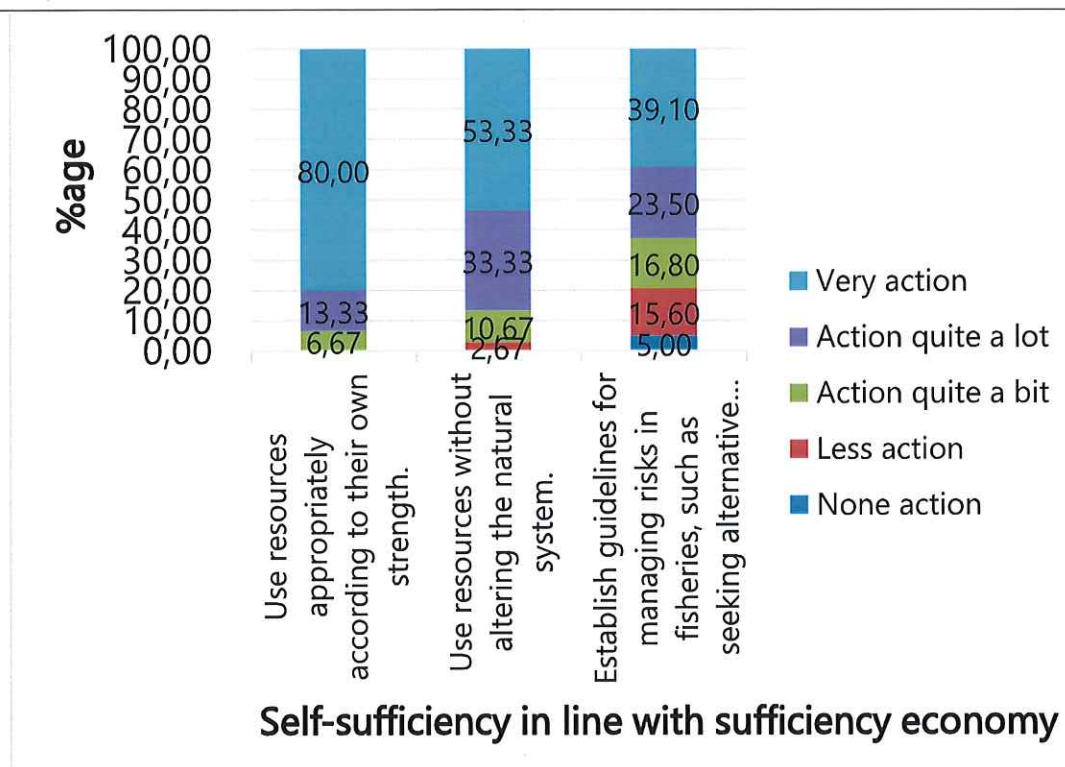


Figure 3. Social Compliance self-sufficiency in line with sufficiency economy

From Table 8, the level of self-responsibility found that fishermen had the highest level of self-responsibility, 54.67%, second only to 32.00% of their responsibilities, and 9.33% of fishermen were low.

Table 8. Level of Responsibility for Fishermen at Kiew Lom Dam Lampang Province towards Fisheries Resource Management. (n=75)

Corporate Social Responsibility Level	Number (case)	Percentage
Low (< 70 per cent)	7	9.33
Moderate (70-80 %)	24	32.00
High (> 80 %)	41	54.67

V. CORPORATE SOCIAL RESPONSIBILITY

From Table 9, if you consider the social responsibility issues that all fishermen have acted on the most, it is the issue of "Encouraging good people to rule the country and control the bad from power" (100.00%), second only to the issue of "adherence to coexistence based on discipline for order in society" 21 (67.00%).

As part of the level of social responsibility It found that the percentage of fishermen's social responsibility scores showed that most fishermen had the highest level of social responsibility (66.67%), second only to low (20.00%) and moderate social responsibility. (13.33%) (Table 10).

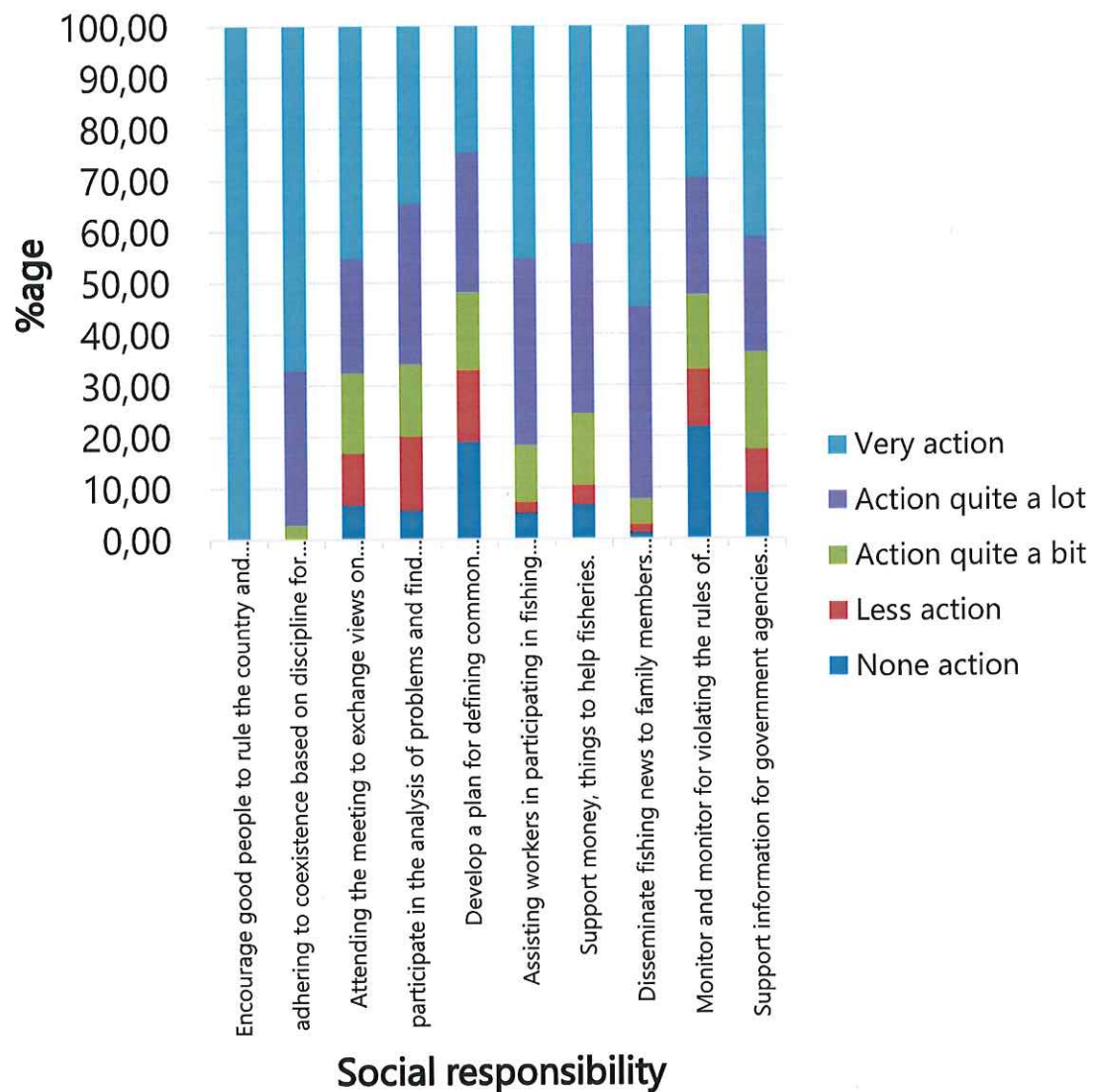


Figure 4. Social responsibility of fishermen at Kiew Lom Dam Lampang Province towards Fisheries Resource Management

Table 10 Level of Fisherman Social Responsibility at Kiew Lom Dam Lampang Province towards Fisheries Resource Management. (n=75)

Corporate Social Responsibility Level	Number (case)	Percentage
Low (< 70 per cent)	15	20.00
Moderate (70-80 %)	10	13.33
High (> 80 %)	50	66.67

Responsibility of fishermen at Kiew Lom Dam Lampang Province towards Overall Fisheries Resource Management

Fishermen have a %age of overall responsibility scores of local fishermen in managing crab resources. There is a high level of image responsibility included (45.3 %), almost a third of fishermen are responsible for the picture included moderately (32.4 %) and another 22.3 % of responsible fishermen are low (Table 11).

Table 11 Level of Responsibility of Fishermen at Kiew Lom Dam Lampang Province towards Fisheries Resource Management (n=75)

Overall responsibility level	Number (case)	Percentage
Low (< 70 per cent)	16	21.33
Moderate (70-80 %)	24	32.00
High (> 80 %)	35	46.67

Factors affecting the responsibility of fishermen at Kiew Lom Dam Lampang Province towards Fisheries Resource Management

It was found that resource knowledge and fisheries resource management measures have a statistically significant impact on overall favorability and social responsibility ($P < 0.05$), community group attendance, and support for fisheries resource management activities from government agencies. These factors impact overall responsibility, responsibility for the episode itself and social responsibility.

Table 12 Factors Affecting Fishermen's Responsibilities at Kiew Lom Dam Lampang Province towards Fisheries Resource Management

Variable	Non-community fishing group meetings		
	knowledge	Attending group meetings in a community	Support for activities to manage the Fisheries assets from government agencies.
Overall responsibility	0.03*	0.00*	0.00*
Self-responsibility	0.16	0.00*	0.01*
Corporate Social Responsibility	0.01*	0.01*	0.00*

VI. SUMMARY AND DISCUSSION

Resource knowledge and fisheries resource management measures have a statistically significant impact on overall responsibility and social responsibility ($P < 0.05$), in line with research conducted by Surachet (1991). This is due to the presence of knowledge subjects to be used in the work. Knowing how it works, knowing people, and knowing, modifying and applying subjects are important elements of expression of appropriate behavior. It is also consistent with Malika (2001), which says that the behavior that a person will display in a work environment may be formal or informal, beneficial or punishable to the organization. It is caused by individual factors, in which competence and skills are variables that influence work behavior. It shows that if fishermen have knowledge of fishing resources and measures to manage fishing resources, they will show local fishermen responsible behavior in managing fisheries

resources. Therefore, local fishermen should be encouraged to have more knowledge of resources and measures to manage fisheries resources so that people are responsible for managing fisheries resources.

Attending group meetings in the community also affects overall responsibility. Suthida (2013) studied factors related to the operational responsibility behavior of Department of Disaster Prevention and Mitigation personnel. The Ministry of Interior found that the interaction between variables aimed at the future and self-control and social transmission of work that affect operational responsibility behavior and support for fisheries resource management activities from government agencies. This also affects overall responsibility as shown by Marisa's research (2005) that studied the psychology and social situations that affect behavior are responsible for the work of tv producers who find that tv producers who receive social support from high-agency personnel are highly responsible for their work. This is in line with Panida's research (2004), that studied psychosocial factors related to the responsible behavior of mechanics in the group. ThaiRungrung Union Car Public Company Limited also found that social support is associated with the same correlation as responsible service work. Therefore, to make fishermen more responsible for managing fisheries resources, there should be added social support, such as the government holding meetings of fishermen groups to provide resource knowledge and measures to manage fisheries resources, or to hold forums for more fishermen to discuss resource management in the area.

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